

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

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FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 | SUB-STRAND 2.5: AREA OF POLYGONS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This Final Explanation document is your guide for answering the driving question in full. Read it carefully before you write your own response. Your goal is to show that you can calculate the area of ANY polygon — including irregular ones like the Kericho community land plot — and explain how your method works. Use the sections below as a framework. Each section has a prompt telling you what to include, and an exemplar showing you the quality and style of thinking expected. Write in your own words; do not copy the exemplars word-for-word. Use diagrams, calculations, and real-context reasoning wherever you can.

Section 1: Restate the Problem and Explain Why It Is Challenging

In your own words, describe the Kericho land plot situation and explain why calculating the area of an irregular polygon is more challenging than calculating the area of a simple shape like a rectangle or triangle. What

The community near Kericho needs to divide an irregularly shaped seven-sided plot of land fairly among three farming families who will use it to grow tea and maize. This is challenging because the plot does not match any single standard formula — it is not a rectangle, a triangle, or a regular hexagon. Each side has a different length, and the interior angles are all unequal. You cannot just multiply length by width. Getting the calculation wrong is not just a maths mistake; it affects people's livelihoods, food security, and trust within the community. That is why we need a reliable method that works for ANY polygon, no matter how irregular it is.

makes this problem matter for real people's lives?

Section 2: The Core Strategy — Decomposition into Known Shapes

Explain the key mathematical strategy you have learned for finding the area of any polygon. What is decomposition? Which shapes do you decompose a polygon into, and why are those shapes useful? Include a labelled sketch showing how you would decompose a polygon (you may use the Kericho plot or a polygon of your own) into triangles and/or rectangles and/or trapezoids.

The core strategy is called DECOMPOSITION — you divide (decompose) any polygon into smaller shapes whose area formulas you already know: triangles, rectangles, and trapezoids. This works because every polygon, no matter how many sides or how irregular, can be split into a set of non-overlapping triangles that together cover the entire shape with no gaps. Triangles are especially powerful because you only need a base and a perpendicular height to find the area: $A = \frac{1}{2} \times \text{base} \times \text{height}$. For the Kericho seven-sided plot, you could draw diagonals from one vertex to all non-adjacent vertices, splitting the polygon into five triangles. You then calculate the area of each triangle separately and add them all together. The total is the area of the whole plot. You can also decompose using rectangles and trapezoids when those shapes appear naturally in the polygon, because their formulas ($A = \text{length} \times \text{width}$ and $A = \frac{1}{2}(a + b) \times h$ respectively) are equally straightforward. Always check that your smaller shapes do not overlap and that they cover the entire polygon.

Section 3: Worked Calculation — Show Your Method Clearly

Using the Kericho land plot (or a clearly labelled irregular polygon with at least five sides), show a complete, step-by-step area calculation. Label all measurements. Show how you decomposed the shape, calculated each sub-area, and summed them. State the final total area with correct units (e.g., m², hectares). Then show how you would divide the total area into three equal shares for the three farming families.

Suppose the Kericho plot has been drawn to scale and measurements have been taken. After decomposing it into five triangles (T1 to T5), the individual areas are calculated as follows:

- T1: base = 18 m, height = 12 m → $A = \frac{1}{2} \times 18 \times 12 = 108 \text{ m}^2$
- T2: base = 22 m, height = 9 m → $A = \frac{1}{2} \times 22 \times 9 = 99 \text{ m}^2$
- T3: base = 15 m, height = 14 m → $A = \frac{1}{2} \times 15 \times 14 = 105 \text{ m}^2$
- T4: base = 20 m, height = 11 m → $A = \frac{1}{2} \times 20 \times 11 = 110 \text{ m}^2$
- T5: base = 16 m, height = 10 m → $A = \frac{1}{2} \times 16 \times 10 = 80 \text{ m}^2$

Total area = $108 + 99 + 105 + 110 + 80 = 502 \text{ m}^2$

To divide fairly among three families:

Each family's share = $502 \div 3 \approx 167.3 \text{ m}^2$

You would then mark the boundary lines on the plot so that each section contains approximately 167.3 m². This could be done by grouping triangles and adjusting one boundary line until the three sections are equal. Always re-check your arithmetic and verify that the sub-areas sum back to the total.

Section 4: Connecting to Other Real Kenyan Contexts

The driving question says this method should help solve 'other real Kenyan land and design problems.' Give at least TWO other specific examples from Kenyan life where calculating the area of an irregular polygon is important. For each example, name the context, describe the polygon shape involved, and explain which area formula or decomposition approach you would use.	Example 1 — Smallholder Farm in the Rift Valley: A maize farmer near Nakuru has an L-shaped shamba (farm) with six sides. She wants to know how much fertiliser to buy; fertiliser is sold per square metre. She can decompose the L-shape into two rectangles, calculate each rectangle's area using $A = l \times w$, and add them. Knowing the exact area prevents her from wasting money on too much fertiliser or reducing her yield by buying too little. Example 2 — School Playground Design in Kisumu: A primary school in Kisumu wants to tile an irregularly shaped courtyard with five sides for a new playground. The school committee needs to know the total area in order to order the correct number of paving tiles. A contractor can decompose the courtyard into a rectangle and a trapezoid, apply $A = l \times w$ and $A = \frac{1}{2}(a + b) \times h$, and sum the results to find the exact number of tiles needed — saving the school budget from costly errors. In both cases, the same decomposition strategy from the Kericho land problem applies directly to a real decision that affects people's money, time, and wellbeing.
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Section 5: Answer the Driving Question Directly	
Now answer the driving question in a clear, well-structured paragraph of at least 8 sentences. Your answer must: (a) state the general method for finding the area of ANY polygon; (b) refer back to the Kericho land plot as your evidence; (c) explain what mathematical tools (formulas) you used; (d) explain how the method enables fair division of land; and (e) reflect on why this mathematical skill matters beyond the classroom.	You can calculate the area of ANY polygon — no matter how many sides it has or how irregular it is — by using the strategy of decomposition: dividing the polygon into triangles, rectangles, or trapezoids whose areas you already know how to find, calculating each sub-area, and summing the results. For the Kericho community land plot, this meant drawing diagonals from one vertex to split the seven-sided shape into five non-overlapping triangles, then applying $A = \frac{1}{2} \times \text{base} \times \text{height}$ to each one. Adding all five triangle areas gave the total area of the plot, which could then be divided by three to find each farming family's equal share. The mathematical tools used throughout were the triangle area formula, the rectangle area formula ($A = l \times w$), and the trapezoid area formula ($A = \frac{1}{2}(a + b) \times h$), all of which combine powerfully through decomposition to handle any polygon. This method makes fair land division possible because it replaces guesswork and argument with a transparent, checkable calculation that every member of the community can verify step by step. Beyond the Kericho plot, the same skill applies to farm planning in the Rift Valley, construction projects in Nairobi, and design problems across Kenya wherever land or surface area must be measured accurately. Mathematics, in this way, is not just an abstract school subject — it is a tool for justice, resource management, and informed decision-making in everyday Kenyan life. Mastering area of polygons means you are equipped to solve real problems that affect real families.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Conceptual Understanding of Decomposition	Clearly and accurately explains decomposition as a universal strategy for any polygon. Correctly identifies which sub-shapes (triangles, rectangles, trapezoids) to use and explains why they are appropriate. Makes explicit connection to the Kericho plot.	Explains decomposition correctly and identifies appropriate sub-shapes. Connection to the Kericho plot is present but could be more specific or detailed.	Shows a partial or vague understanding of decomposition. May confuse sub-shapes or apply the strategy only to simple/familiar polygons. Connection to the Kericho context is weak or missing.

Accuracy and Completeness of Area Calculations	All area formulas are stated and applied correctly (triangle: $\frac{1}{2}bh$; rectangle: lw ; trapezoid: $\frac{1}{2}(a+b)h$). All sub-areas are calculated without arithmetic errors. Sub-areas are correctly summed to give the total. Units (m^2 , hectares) are used consistently and correctly throughout.	Area formulas are mostly correct and applied appropriately. Minor arithmetic errors may be present but do not undermine the overall method. Units are included but may be inconsistent in one or two places.	One or more area formulas are missing or incorrectly applied. Significant arithmetic errors are present. Units are missing or incorrectly applied. The final total area cannot be verified from the working shown.
Application to Fair Land Division and Real Kenyan Contexts	Clearly shows how the total area is divided equally among the three families with correct arithmetic. Provides two or more specific, well-explained Kenyan examples (e.g., Rift Valley farm, Kisumu courtyard) that demonstrate transfer of the decomposition strategy to new contexts.	Shows how the total area is divided among three families. Provides at least one Kenyan example with reasonable explanation of how decomposition applies. Examples are relevant but may lack specificity.	Division of land among families is attempted but contains errors or is not clearly explained. Kenyan examples are absent, superficial, or do not demonstrate use of the area strategy.
Use of Mathematical Language, Notation, and Diagrams	Consistently uses precise mathematical vocabulary (decomposition, polygon, vertex, diagonal, perpendicular height, sub-area, etc.). All diagrams are labelled with measurements, sub-shapes are clearly marked, and notation is consistent with Grade 10 expectations.	Uses appropriate mathematical vocabulary most of the time. Diagram is present and mostly labelled, though some labels or measurements may be missing. Notation is generally correct.	Mathematical vocabulary is limited, imprecise, or frequently used incorrectly. Diagram is absent, unlabelled, or does not accurately represent the decomposition. Notation is inconsistent or missing.
Quality of Driving Question Response and Real-World Reasoning	The driving question is answered directly and comprehensively in a well-structured paragraph of 8+ sentences. All five required elements are addressed (general method, Kericho evidence, formulas used, fair division, real-world relevance). The response reflects genuine insight into why mathematical precision matters for people's livelihoods and community decision-making.	The driving question is answered clearly and covers most of the five required elements. The paragraph is adequately developed (6–7 sentences). Real-world reasoning is present but may be somewhat general rather than deeply reflective.	The driving question response is brief, incomplete, or does not directly address the question. Fewer than four of the five required elements are present. Real-world reasoning is absent or limited to a single generic statement.